# From raw data to results you can trust

## Programming & Data Management, Part II (Python) · Lecture 1

Magnus Lodefalk
*Örebro University*

NA446A, Örebro University

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# Today's goals

By the end of today you can:

1. Open a Python project in VS Code that has its own environment, and run a script
2. Organise a project so that someone else can rerun it
3. Load a dataset in pandas and inspect it: size, first rows, summary statistics

# From last time

Answer on your own first, then compare with a neighbour

1. In Stata, how do you load a dataset and look at its first five rows?
2. Where on your computer is the file you worked on last week?

# From last time

Answer on your own first, then compare with a neighbour

1. In Stata, how do you load a dataset and look at its first five rows?
   Answer: `use auto, clear` and `list in 1/5`
2. Where on your computer is the file you worked on last week?
   Answer: If that took more than five seconds, today's second goal is for you

# How Part II works

- Four sessions, each the same three steps: a short lecture, a lab, an exercise sheet
- Exercises go from easy to hard; solutions follow three days later
- Everything is on the course site: `magnuslodefalk.com/na446a`
- The written exam tests the course's learning outcomes; the labs are where you practise them

**Today's lab:** install, open the project, run your first script. Help is in the room.

# One recipe, four sessions




- Every session adds one step. Today: set up the project, then load and inspect.

# Why it matters

# A famous result no one could check

- Reinhart and Rogoff (2010, *AER*): growth falls sharply once public debt exceeds 90% of GDP. Policymakers quoted it in the austerity debate
- 2013: a graduate student, Thomas Herndon, tries to replicate it as homework and cannot
- With the authors' spreadsheet, he finds that the key average used 15 of the 20 countries, plus excluded data and unusual weights

- Published: average growth above 90% debt was $-0.1\%$.

# Corrected, what was growth above 90% debt?

Your bet first: hands up. The answer is on the next slide.

A. About $-0.1\%$, as published
B. About $1\%$
C. About $2\%$

# Corrected: 2.2% growth, not −0.1%

- The spreadsheet left countries out of the key average; the correction puts them back
- The authors accepted the spreadsheet error but disputed how much it mattered

- Either way, no one could check the result until the spreadsheet was shared.

Herndon, Ash and Pollin (2013), PERI Working Paper 322; BBC News, 20 April 2013.

# Which regression produced the key result?

Questions every empirical economist has asked at 11 pm:

- Where is the original raw data file?
- How did I define that control variable two years ago?
- Why do the summary statistics look so strange for the outcome?
- Which of the regressions produced the result in the paper?

All four have the same cure: a project anyone can rerun from raw data to results.

# Seven principles for credible empirical work

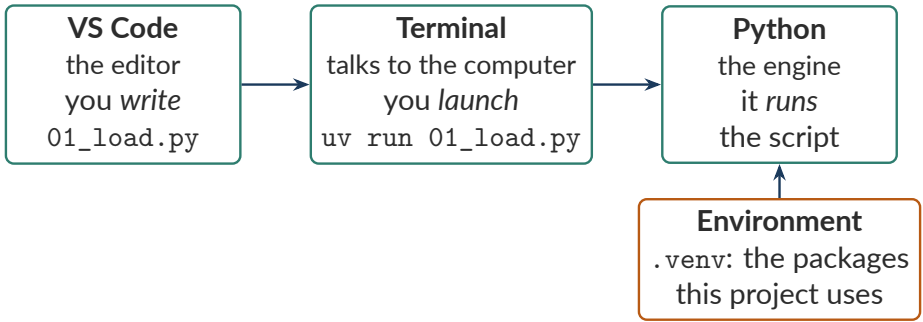
Today we practise the three in teal

1. **Credibility:** results survive scrutiny; log choices, test robustness
2. **Reproducibility:** same results on rerun; one run-all script
3. **Efficiency:** automate, reuse, refactor
4. **Transparency:** a legible path from data to results
5. **Data integrity:** raw data is read-only; check early
6. **Organisation:** one folder layout: raw, code, modified, output
7. **Version control:** a history you can trust

Consistent with the Data and Code Availability Standard that the AEA follows.

# The tools

# Four tools, one chain




One VS Code window shows all three. In Stata, editor and engine are one program.

– "ModuleNotFoundError" almost always means the terminal is not using the project's environment.

# One folder layout, every project

| | |
|---|---|
| `data/raw/` | the original data, never edited |
| `data/processed/` | data your code made |
| `scripts/` | scripts, numbered in the order they run |
| `output/` | tables, figures and logs |
| `README.md` | how to rerun the project |
| `pyproject.toml` | the package versions |
| `run_all.py` | runs every script, in order |

- Delete `data/processed/` and `output/`, run `run_all.py`: everything comes back. That is the test of a good project.

# Check your understanding

Answer on your own first, then compare with a neighbour

You drop three outliers and save the result with `cars.to_csv("data/raw/auto.csv")`. What went wrong?

# Check your understanding

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You drop three outliers and save the result with `cars.to_csv("data/raw/auto.csv")`. What went wrong?

**Answer:** You overwrote the raw data, so the outliers are gone for good. Raw data is read-only: save to `data/processed/auto_clean.csv` instead.

# First steps in Python

# From Stata commands to Python objects

**Stata:** one dataset in memory; commands act on it

```
use auto, clear
summarize mpg
```

**Python:** each dataset is an object; you ask it to act

```
cars = pd.read_csv(path)
cars["mpg"].describe()
```

- You can have many datasets open at once, each with its own name: `cars`, `wages`, `firms`.

# Now I’ll show you in VS Code

# Step 1: load the data and check its size

```
from pathlib import Path
import pandas as pd

RAW = Path("data/raw")
cars = pd.read_csv(RAW / "auto.csv")
cars.shape
```

```
(392, 9)
```

`RAW` points to the raw-data folder; `read_csv` reads the file into a DataFrame; `shape` gives (rows, columns).

- 392 cars and 9 variables. Check the size every time you load data: it is the first sign that something went wrong.

# Step 2: look at the first rows

```
cars[["name", "mpg", "weight", "origin"]].head(3)
```

```
                        name   mpg  weight  origin
0  chevrolet chevelle malibu  18.0    3504       1
1          buick skylark 320  15.0    3693       1
2         plymouth satellite  18.0    3436       1
```

- The left column is the row index, starting at 0, not 1. Python counts from zero.

# Step 3: summary statistics

```
cars[["mpg", "weight"]].describe().round(1)
```

```
          mpg  weight
count   392.0   392.0
mean     23.4  2977.6
std       7.8   849.4
min       9.0  1613.0
50%      22.8  2803.5
max      46.6  5140.0
```

- Same numbers as Stata's `summarize`; the median (50%) comes for free.

Rows 25% and 75% omitted here to fit the slide.

# Check your understanding

Answer on your own first, then compare with a neighbour

Predict the result before you run it: `cars.groupby("origin")["mpg"].mean()`. Which origin do you expect to have the highest mean mpg, and how would you write it in Stata?

**Now I'll show you in VS Code**

# Check your understanding

Answer on your own first, then compare with a neighbour

Predict the result before you run it: `cars.groupby("origin")["mpg"].mean()`. Which origin do you expect to have the highest mean mpg, and how would you write it in Stata?

**Now I'll show you in VS Code**

**Answer:** Origin 3 (Japan), 30.5 mpg, against 20.0 for the US and 27.6 for Europe. In Stata: `tabstat mpg, by(origin)`.

# Five rules for using AI in this course

The same rules in every part of the course

1. **Try first yourself:** write in a comment what the code should do
2. **Use AI to explain, debug or suggest an alternative**, not to write the whole script
3. **Test** what it gives you on a case where you know the answer
4. **Explain** every line, and where the code stops working
5. **Own the result:** mark lines that came from AI; you answer for all of them

- Why: in a trial with about 1,000 students, ChatGPT raised practice grades by 48% but lowered grades on a later exam without AI by 17%, and the students did not notice (Bastani et al., 2025, *PNAS*).

# Today's lab

1. Install the tools and open the course project in VS Code *goal 1*
2. Set up the four folders; run today's Steps 1–3 as one script *goal 2*
3. Fix an AI-written script: `lab01_fix_the_bug.py` runs without an error but gives a wrong answer. Find it by testing a case you know *goal 3*

– Step 3 practises rules 3 and 4: code that runs is not code that is right.

Stuck? Paste the course tutor prompt (course site, AI page) into your chatbot first: it gives hints, not answers.

# Summary

1. Each project has its own environment; the terminal must use it
2. Raw data is read-only; code writes only to `data/processed/` and `output/`
3. In pandas each dataset is an object: `shape`, `head()`, `describe()`

**Before next time:** read the Turrell chapter (the parts named on the Readings page); try the self-test without AI.

# Test yourself before next time

Without AI: this is what you should manage on your own

1. Draw the chain from VS Code to the Python interpreter. Where do packages live?
2. Why is `data/raw/` read-only, and where does cleaned data go?
3. What does `cars.shape` return, and why check it after every load?
4. Write the pandas line that gives the mean weight by origin.

# Appendix

# When the setup fails

- **“ModuleNotFoundError: pandas”:** the terminal is not in the project’s environment. Activate it, or pick the interpreter ending in `.venv` in VS Code
- **“File not found”:** VS Code is open on the wrong folder. Open the project folder itself, and use paths relative to it
- **Nothing works:** delete the `.venv` folder and type `uv sync` again; for today, work with a neighbour and email me

# Stata and Python terms

| Python / data science | Stata |
|---|---|
| DataFrame | the dataset in memory |
| row, record, instance | observation |
| column, feature | variable |
| target | dependent variable |
| missing value `<NA>`, `NaN` | . |
| method: `cars.describe()` | command: `summarize` |
| package: `pip install` | `ssc install` |

# References and further reading


- Alexander, R. (2013). Reinhart, Rogoff… and Herndon: The student who caught out the profs. *BBC News*, 20 April.
- Bastani, H., O. Bastani, A. Sungu, H. Ge, Ö. Kabakcı and R. Mariman (2025). Generative AI without guardrails can harm learning: Evidence from high school mathematics. *PNAS* 122(26): e2422633122.
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